Python Crash Course: How the Code Works

Introduction

In this worksheet, you will learn how Python code actually runs, step by step.

This worksheet focuses on small changes to code and what those changes do. Each section introduces one idea at a time, with short examples you can run and modify. The goal is not to memorize commands, but to understand what Python is doing and why.

You will practice:

- Running code and reading output
- Changing numbers and words in code
- Fixing simple errors
- Repeating code with loops
- Making decisions with if statements
- Using functions to organize code

Move through the sections in order. Try running every example before changing it. If something breaks, that is expected. Errors are part of learning how code works.

By the end, you should feel more comfortable reading code, changing it, and figuring out what it does on your own.

Section 1: Running code and Reading Output

Goal: Understand that Python runs top to bottom, and prints out exactly what you tell it to.

```
print("Hello")
print("World")
```

Task 1

- Run this
- Write down or type (in a Google doc) exactly what it prints

Task 2

Change the code so that it prints:

World

Hello

Section 2: Variables (Names for Things)

A variable is a name that stores a values

```
x = 5
print(x)
```

Task 2

Change the x so the output is 10

```
name = "Alex"
print(name)
```

Task 3

Change the name to your own

```
x = 5
y = 3
print(x + y)
```

Task 4

Change the numbers so the output is 20

Section 3: Strings vs Numbers (This is important)

```
print(5 + 5)
print("5" + "5")
```

Task 5

- Run this code
- Write down or type what each line prints
- Pay attention to whether there are “” around the numbers or not

```
print("5" + 5)
```

Task 6

- Run this
- What happens?

This is called an **error.** Errors tell you what Python did not understand

Section 4: Fixing Errors

```
print("Hello)
```

Task 7

- Run this (pay attention to the quotations)
- Read through the red error message
- Fix the code so it runs

```
print(hello)
```

Task 8

- Run this
- Fix it so it prints the word hello

Section 5: Repeating Code (Loops)

A loop repeats code multiple times.

Task 9

- Run the code below
- Write down what it prints

```
for i in range(3):
    print("Hi")
```

Task 10

Change the 3 so that "Hi" prints 5 times

Task 11

- Run the code below
- Write down the numbers that print

```
for i in range(4):
    print(i)
```

Task 12
Change the code below so that it prints:
2
3
4
5
6

```
for i in range(1, 6):
    print(i)
```

Section 6: Making Decisions (if Statements)

An if statement checks whether something is true.

Task 13
Run the code below

```
x = 10

if x > 5:
    print("Big")
```

Change x so that nothing prints

Section 7: Functions (Reusable Code)

A function is a block of code that you can run again.

Task 15
Run the code below

```
def say_hi():
    print("Hi")

say_hi()
```

Change the function so that it prints Hello instead

Task 16
Run the code below

```
def say_name(name):
    print(name)

say_name("Alex")
```

Change "Alex" to your own name

Task 17
Run the code below

```
def add(a, b):
    print(a + b)

add(2, 3)
```

Change the numbers so that it prints 10

Section 8: Putting It All Together

This example uses a function and a loop together.

Task 18
Run the code below

```
def repeat_word(word, times):
    for i in range(times):
        print(word)

repeat_word("Hi", 3)
```

Change the code so that it prints:
Hello
Hello
Hello
Hello

Final Notes

You do not need to memorize any of this.

The goal is to be able to:
- Read code
- Change small parts of code
- Understand what changed and why
- Fix simple errors when they happen

Once you are finished, you can continue on to Python Crash Course: Building Simple Programs! ☺